

Session2-Exercises-solution.R

```
#####  
### Session 2 Exercises ###  
#####  
  
#2.1  
# Load the tidyverse  
library(tidyverse)  
  
# Use the read_delim function to Look at the file with a very full syntax:  
stoppages <- read_delim(file='workstoppages.txt', delim='^')  
  
## Rows: 434 Columns: 3  
## — Column specification —————  
## Delimiter: "\""   
## chr (1): period  
## dbl (2): year, workers  
##  
## i Use `spec()` to retrieve the full column specification for this data.  
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.  
  
glimpse(stoppages)  
  
## Rows: 434  
## Columns: 3  
## $ year    <dbl> 1981, 1981, 1981, 1981, 1981, 1981, 1981, 1981, 1981, 1981, 19...  
## $ period  <chr> "January", "February", "March", "April", "May", "June", "July"...  
## $ workers <dbl> 29600, 20900, 207800, 223500, 259000, 415100, 125400, 86600, 6...  
  
#2.2  
# Try just reading the file without other arguments  
library(readxl)  
breakfast <- read_excel('breakfast.xlsx')  
  
## New names:  
## • `` -> `...2`  
## • `` -> `...3`  
## • `` -> `...4`  
## • `` -> `...5`  
## • `` -> `...6`  
## • `` -> `...7`
```

```
glimpse(breakfast)
```

```
## Rows: 46
## Columns: 7
## $ `School Breakfast Program Participation and Meals Served` <chr> "(Data as of..."
## $ ...2 <chr> NA, "    -..."
## $ ...3 <chr> NA, NA, "Red..."
## $ ...4 <chr> NA, NA, "Pai..."
## $ ...5 <chr> NA, NA, "Tot..."
## $ ...6 <chr> NA, NA, "Mea..."
## $ ...7 <chr> NA, NA, "Fre..."
```

Try skipping three lines

```
breakfast <- read_excel('breakfast.xlsx', skip=3)
glimpse(breakfast)
```

```
## Rows: 43
## Columns: 7
## $ `Fiscal Years` <chr> "    ", "1975", "1976", "1977", "1978", "19..."
## $ Free <chr> "-----Millions-----"
## $ `Red. Price` <dbl> NA, 0.04, 0.06, 0.11, 0.16, 0.21, 0.25, 0.2...
## $ Paid <dbl> NA, 0.33, 0.37, 0.36, 0.42, 0.54, 0.56, 0.5...
## $ `Total` <dbl> NA, 1.82, 2.20, 2.49, 2.80, 3.32, 3.60, 3.8...
## $ `Meals Served` <dbl> NA, 294.70, 353.60, 434.30, 478.80, 565.60,...
## $ `Free/ RP of Total Meals` <chr> "--Percent--", "82.1", "84.2", "85.7", "85..."
```

It looks like we need to skip five lines, which will remove the column names So let's create a vector with column names

```
names <- c("Year", "FreeStudents", "ReducedStudents", "PaidStudents",
"TotalStudents", "MealsServed", "PercentFree")
```

And then try reading the file again

```
breakfast <- read_excel('breakfast.xlsx', skip=5, col_names = names)
glimpse(breakfast)
```

```
## Rows: 42
## Columns: 7
## $ Year <dbl> 1975, 1976, 1977, 1978, 1979, 1980, 1981, 1982, 1983, ...
## $ FreeStudents <dbl> 1.45, 1.76, 2.02, 2.23, 2.56, 2.79, 3.05, 2.80, 2.87, ...
## $ ReducedStudents <dbl> 0.04, 0.06, 0.11, 0.16, 0.21, 0.25, 0.25, 0.16, 0.15, ...
## $ PaidStudents <dbl> 0.33, 0.37, 0.36, 0.42, 0.54, 0.56, 0.51, 0.36, 0.34, ...
## $ TotalStudents <dbl> 1.82, 2.20, 2.49, 2.80, 3.32, 3.60, 3.81, 3.32, 3.36, ...
## $ MealsServed <dbl> 294.70, 353.60, 434.30, 478.80, 565.60, 619.90, 644.20...
## $ PercentFree <dbl> 82.1, 84.2, 85.7, 85.3, 84.1, 85.2, 86.9, 89.3, 90.3, ...
```

We can do a little quick manipulation of this tibble.

First, convert the numbers of students and meals to real values

```
breakfast <- mutate(breakfast, FreeStudents=FreeStudents*1000000,
  ReducedStudents=ReducedStudents * 1000000,
  PaidStudents = PaidStudents * 1000000,
  TotalStudents = TotalStudents * 1000000,
  MealsServed = MealsServed * 1000000,
  PercentFree = PercentFree/100)
```

```

#After Learning about the pipe
#breakfast <- breakfast %>% mutate(FreeStudents=FreeStudents*1000000,
#      ReducedStudents=ReducedStudents * 1000000,
#      PaidStudents = PaidStudents * 1000000,
#      TotalStudents = TotalStudents * 1000000,
#      MealsServed = MealsServed * 1000000,
#      PercentFree = PercentFree/100)

```

```
glimpse(breakfast)
```

```

## Rows: 42
## Columns: 7
## $ Year      <dbl> 1975, 1976, 1977, 1978, 1979, 1980, 1981, 1982, 1983, ...
## $ FreeStudents <dbl> 1450000, 1760000, 2020000, 2230000, 2560000, 2790000, ...
## $ ReducedStudents <dbl> 40000, 60000, 110000, 160000, 210000, 250000, 250000, ...
## $ PaidStudents <dbl> 330000, 370000, 360000, 420000, 540000, 560000, 510000...
## $ TotalStudents <dbl> 1820000, 2200000, 2490000, 2800000, 3320000, 3600000, ...
## $ MealsServed <dbl> 294700000, 353600000, 434300000, 478800000, 565600000,...
## $ PercentFree <dbl> 0.821, 0.842, 0.857, 0.853, 0.841, 0.852, 0.869, 0.893...

```

#2.3

#a-b

```
read_csv("employees_broken.txt")
```

```

## # A tibble: 7 × 1
##   `EMPLOYEE DATA EXPORT - INTERNAL USE ONLY`
##   <chr>
## 1 Generated on: 2023-11-15
## 2 -----
## 3 001John Smith    19850423M  45000
## 4 002Alice Brown  19911205F  52000
## 5 003Mark Davis   19790612M
## 6 004SophieMartin 19880228F  61000
## 7 005Paul Durand  19930517 M 48000

```

*# It seems that the first three lines are just headers and/or metadata
No clear delimiter, but it seems to be fixed width.*

#c

```

employees_raw <- read_fwf(
  "employees_broken.txt",
  fwf_positions(
    start = c(1, 4, 17, 25, 28),
    end   = c(3, 16, 24, 26, 33),
    col_names = c("id", "name", "birthdate", "gender", "salary")
  ),
  skip = 3,
  na = c("", " ")
)

```

employees_raw

```
## # A tibble: 5 × 5
##   id    name      birthdate gender salary
##   <chr> <chr>      <dbl> <chr>   <dbl>
## 1 001    John Smith   19850423 M      45000
## 2 002    Alice Brown  19911205 F      52000
## 3 003    Mark Davis   19790612 M      NA
## 4 004    SophieMartin 19880228 F      61000
## 5 005    Paul Durand  19930517 M      48000
```

#d

Name is in one column, should be separated into first and last name
Birthdate is in numeric format, it should be date
Salary has a missing value

#e

```
read_csv("employees_broken2.txt")
```

```
## # A tibble: 7 × 1
##   `EMPLOYEE DATA EXPORT - INTERNAL USE ONLY`
##   <chr>
## 1 Generated on: 2023-11-15
## 2 -----
## 3 001John Smith   19850423M  45000
## 4 002Alice Brown  19911205F  52000
## 5 003Mark Davis   19790612M
## 6 004SophieMartin19880228F  61000
## 7 005Paul Durand  19930517 M 48000
```

It is a seemingly fixed width file, but in fact, at least SophieMartin's record is different. It can be solved either manually or reading the file line-by-line as an ETL solution. You cannot “fix” this with clever column positions.

Many solutions will be studied later :-)

Observed invariant rules

From inspection:

- # • id = first 3 digits*
- # • birthdate = exactly 8 consecutive digits*
- # • gender = single letter M or F*
- # • salary = trailing digits (optional)*
- # • name = whatever remains between id and birthdate*

These rules hold even when spacing shifts.

The meta-lesson you take away

- # - Data formats are contracts, not guarantees*
- # - Tools enforce assumptions – they do not verify them*
- # - The analyst is responsible for detecting violations*
- # - Sometimes the “right” solution is to abandon the format itself*

These are professional-grade insights, even if the code remains simple.

#2.4

#a

```
library(readxl)
```

```
planets <- read_excel('planets.xlsx', sheet=2)
```

#b

*# The data is wrongly structured, thereby the necessary data
is in the first row, except the first column*

```
mean(as.numeric(planets[1, -1]))
```

```
## [1] 296.2918
```

#2.5

```
#install.packages("haven")
```

```
library(haven)
```

#now you can read the data file

```
spss_table <- read_spss('database test.sav')
```

#check it

```
spss_table
```

```
## # A tibble: 50 × 7
```

```
##   Observation Birthyearofthehouseho...1 Averageamountspentper...2 Nbofvisitspermonth
##   <dbl>          <dbl>          <dbl>          <dbl>
## 1           1          1976          382.           3
## 2           2          1991          361.           4
## 3           3          1956          325.           5
## 4           4          1979          304.           2
## 5           5          1978          457.           3
## 6           6          1961          447            2
## 7           7          1964          295.           6
## 8           8          1977          386.           6
## 9           9          1977          385.           3
## 10          10          1986          827.           3
```

```
## # i 40 more rows
```

```
## # i abbreviated names: 1Birthyearofthehouseholder, 2Averageamountspentpermonth
```

```
## # i 3 more variables: Monthlyincomeofthehousehold <dbl>,
```

```
## #   Maincriterionofpurchase <chr>, Nbofchildren <dbl>
```

#glimpse it

```
glimpse(spss_table)
```

```
## Rows: 50
```

```
## Columns: 7
```

```
## $ Observation      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, ...
## $ Birthyearofthehouseholder <dbl> 1976, 1991, 1956, 1979, 1978, 1961, 1964, ...
## $ Averageamountspentpermonth <dbl> 382.1, 361.4, 324.6, 303.7, 456.7, 447.0, ...
## $ Nbofvisitspermonth <dbl> 3, 4, 5, 2, 3, 2, 6, 6, 3, 3, 4, 6, 3, 7, ...
## $ Monthlyincomeofthehousehold <dbl> 2376, 2564, 1549, 1986, 1340, 2014, 3248, ...
## $ Maincriterionofpurchase <chr> "lowest price", "known brand", "known bran...
## $ Nbofchildren      <dbl> 0, 2, 1, 2, 0, 1, 0, 1, 0, 3, 0, 1, 1, 0, ...
```

#and see the difference between tibble and data frame

```
spss_table1 <- as.data.frame(spss_table)
```

```
spss_table1
```

##	Observation	Birthyearofthehouseholder	Averageamountspentpermonth
## 1	1	1976	382.1
## 2	2	1991	361.4
## 3	3	1956	324.6
## 4	4	1979	303.7
## 5	5	1978	456.7
## 6	6	1961	447.0
## 7	7	1964	294.8
## 8	8	1977	386.3
## 9	9	1977	385.2
## 10	10	1986	827.4
## 11	11	1990	457.2
## 12	12	1986	470.8
## 13	13	1959	379.7
## 14	14	1973	295.1
## 15	15	1978	424.1
## 16	16	1967	349.5
## 17	17	1991	491.5
## 18	18	1965	308.1
## 19	19	1992	402.7
## 20	20	1986	333.6
## 21	21	1991	425.0
## 22	22	1987	295.3
## 23	23	1971	379.3
## 24	24	1993	195.7
## 25	25	1964	376.8
## 26	26	1986	432.8
## 27	27	1966	515.1
## 28	28	1994	306.4
## 29	29	1989	288.0
## 30	30	1960	294.4
## 31	31	1971	273.3
## 32	32	1985	415.3
## 33	33	1994	365.8
## 34	34	1972	454.1
## 35	35	1963	145.6
## 36	36	1991	368.3
## 37	37	1969	471.1
## 38	38	1997	876.2
## 39	39	1970	409.2
## 40	40	1989	420.7
## 41	41	1969	279.2
## 42	42	1982	299.1
## 43	43	1981	411.3
## 44	44	1995	454.7
## 45	45	1997	471.8
## 46	46	1990	419.8
## 47	47	1988	488.3
## 48	48	1977	352.6
## 49	49	0	507.9
## 50	50	0	420.2

##	Nbofvisitspermonth	Monthlyincomeofthehousehold	Maincriterionofpurchase
## 1	3	2376	lowest price
## 2	4	2564	known brand
## 3	5	1549	known brand
## 4	2	1986	origin of products
## 5	3	1340	habits
## 6	2	2014	known brand
## 7	6	3248	lowest price
## 8	6	1608	lowest price
## 9	3	2180	lowest price
## 10	3	2448	lowest price
## 11	4	1877	product composition
## 12	6	4063	lowest price
## 13	3	2536	lowest price
## 14	7	3131	lowest price
## 15	4	1925	habits
## 16	2	1961	known brand
## 17	3	1685	known brand
## 18	4	3526	lowest price
## 19	5	4007	lowest price
## 20	7	3545	known brand
## 21	8	4365	lowest price
## 22	8	4537	origin of products
## 23	5	3414	lowest price
## 24	6	2392	lowest price
## 25	5	2728	known brand
## 26	4	4440	lowest price
## 27	3	3110	origin of products
## 28	4	3590	lowest price
## 29	4	3288	known brand
## 30	4	2957	lowest price
## 31	4	2899	lowest price
## 32	3	4261	known brand
## 33	7	1536	known brand
## 34	6	4236	lowest price
## 35	4	1604	lowest price
## 36	5	2087	known brand
## 37	2	3469	known brand
## 38	6	3466	origin of products
## 39	1	2209	lowest price
## 40	7	2942	origin of products
## 41	7	1996	habits
## 42	6	1938	known brand
## 43	3	3219	origin of products
## 44	1	1672	lowest price
## 45	3	2806	known brand
## 46	3	2285	lowest price
## 47	5	3092	lowest price
## 48	5	1789	lowest price
## 49	7	2622	lowest price
## 50	3	2253	product composition
##	Nbofchildren		
## 1	0		
## 2	2		
## 3	1		

## 4	2
## 5	0
## 6	1
## 7	0
## 8	1
## 9	0
## 10	3
## 11	0
## 12	1
## 13	1
## 14	0
## 15	0
## 16	1
## 17	2
## 18	1
## 19	1
## 20	1
## 21	2
## 22	2
## 23	0
## 24	0
## 25	1
## 26	0
## 27	0
## 28	1
## 29	2
## 30	4
## 31	0
## 32	0
## 33	1
## 34	1
## 35	1
## 36	1
## 37	1
## 38	4
## 39	1
## 40	1
## 41	1
## 42	2
## 43	0
## 44	2
## 45	1
## 46	0
## 47	1
## 48	2
## 49	1
## 50	1

#2.6

```
library(readxl)
names <- c("ClinicName", "ClinicLocation", "Neighbourhood",
           "Address", "ContactNumber", "OperationalHours",
           "Services")
toronto <- read_excel("toronto project.xlsx", 2,
                     skip = 4, col_names = names)
```

#b

```
toronto <- toronto[, -3]
```

#2.7

#a

```
library(haven)
dogs <- read_dta("dogs.dta")
```

#b

```
glimpse(dogs)
```

```
## Rows: 48
## Columns: 6
## $ NameLatin    <chr> "Albina", "Bars / Chaika", "Belka", "Belyanka", "Bulba", "...
## $ NameEnglish  <chr> "Whitey", "Panther / Seagull", "Squirrel", "Little Whitey"...
## $ Gender       <chr> "Female", "Female", "Female", "Female", "Female", "Female"...
## $ Flights      <chr> "1956-05-14,1956-06-07,1956-06-14", "1960-07-28", "1957-08...
## $ Fate         <chr> "Survived", "Died 1960-07-28", "Survived", "Survived", "Di...
## $ Notes        <chr> " ", " ", " ", " ", " ", " ", " ", " ", " ", " ", " ", " ", " ", " "...
```